

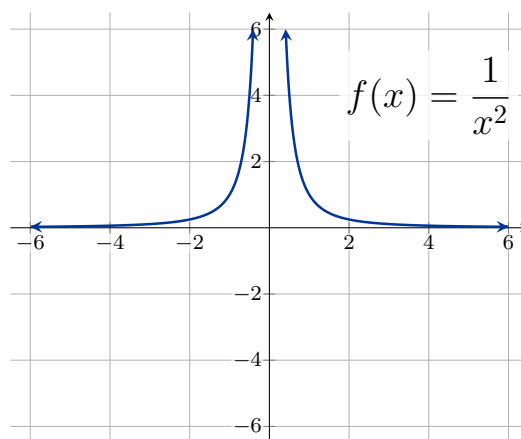
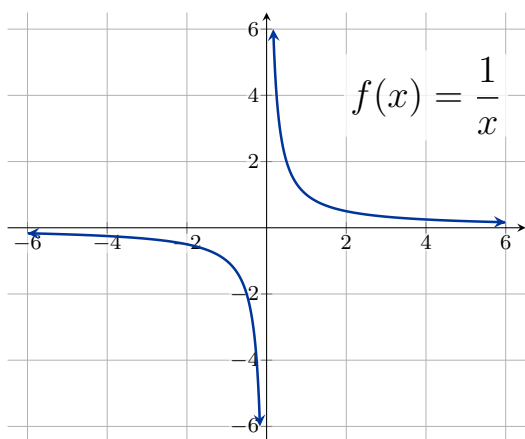
3.7: Rational Functions

Definition. (Rational Function)

A **rational function** is a function of the form

$$r(x) = \frac{f(x)}{g(x)},$$

where $f(x)$ and $g(x)$ are polynomials.



Definition. (Asymptotes)

A **vertical asymptote** of a graph is a vertical line $x = a$ where the graph tends towards positive or negative infinity as the input approaches a from either the left or the right. We write

$$\text{As } x \rightarrow a^-, f(x) \rightarrow \pm\infty \quad \text{or} \quad x \rightarrow a^+, f(x) \rightarrow \pm\infty.$$

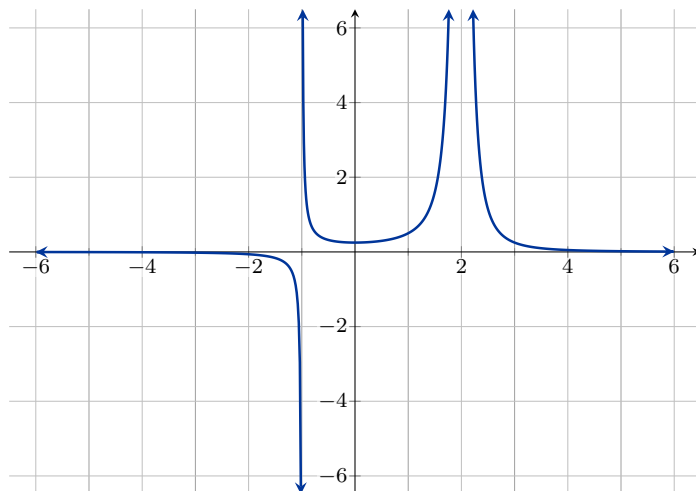
A **horizontal asymptote** of a graph is a horizontal line $y = b$ where the graph approaches the line as the inputs increase or decrease without bound. We write

$$\text{As } x \rightarrow -\infty \text{ or } x \rightarrow \infty, f(x) \rightarrow b.$$

Example. Describe the asymptotes, intercepts, and domain of

$$f(x) = \frac{1}{(x+1)(x-2)^2}$$

using the graph below:



$$x \rightarrow -1^-, f(x) \rightarrow -\infty \qquad x \rightarrow 2^-, f(x) \rightarrow \infty$$

$$x \rightarrow -1^+, f(x) \rightarrow \infty \qquad x \rightarrow 2^+, f(x) \rightarrow \infty$$

Vertical asymptotes: $x = -1, x = 2$

$$\frac{1}{(x+1)(x-2)^2} = \frac{1}{x^3 - 3x^2 + 4}$$

$\leftarrow$ Degree 0
 $\leftarrow$ Degree 3

Horizontal asymptote: $y = 0$ $\leftarrow$

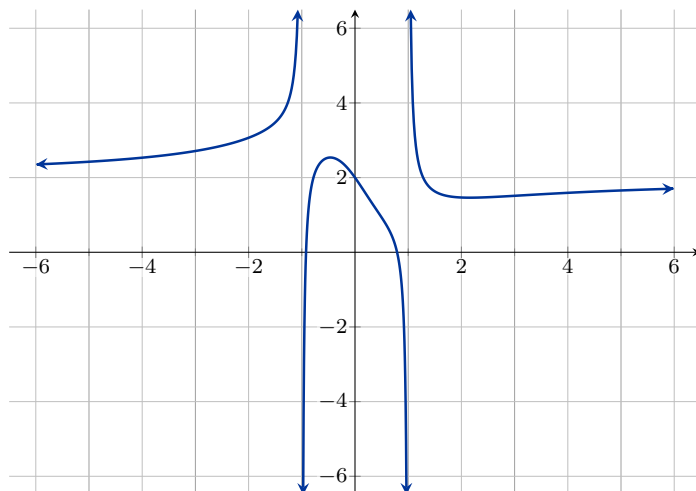
When the degree of the numerator is higher than the degree of the denominator, the horizontal asymptote is $y=0$

Domain: $(-\infty, -1) \cup (-1, 2) \cup (2, \infty)$

Example. Describe the asymptotes, intercepts, and domain of

$$f(x) = \frac{2x^4 - 2x^3 + x^2 + 2x - 2}{(x^2 - 1)(x^2 + 1)} = \frac{2x^4 - 2x^3 + x^2 + 2x - 2}{x^4 - 1}$$

using the graph below:



$$x \rightarrow -1^-, f(x) \rightarrow \infty$$

$$x \rightarrow 1^-, f(x) \rightarrow -\infty$$

$$x \rightarrow -1^+, f(x) \rightarrow -\infty$$

$$x \rightarrow 1^+, f(x) \rightarrow \infty$$

Vertical asymptotes: $x = -1, x = 1$

$$\frac{2x^4 - 2x^3 + x^2 + 2x - 2}{x^4 - 1} \left(\frac{1/x^4}{1/x^4} \right) = \frac{2 - \frac{2}{x} + \frac{1}{x^2} + \frac{2}{x^3} - \frac{2}{x^4}}{1 - \frac{1}{x^4}}$$

When the degree of the numerator is equal to the degree of the denominator, the horizontal asymptote is the ratio of leading terms

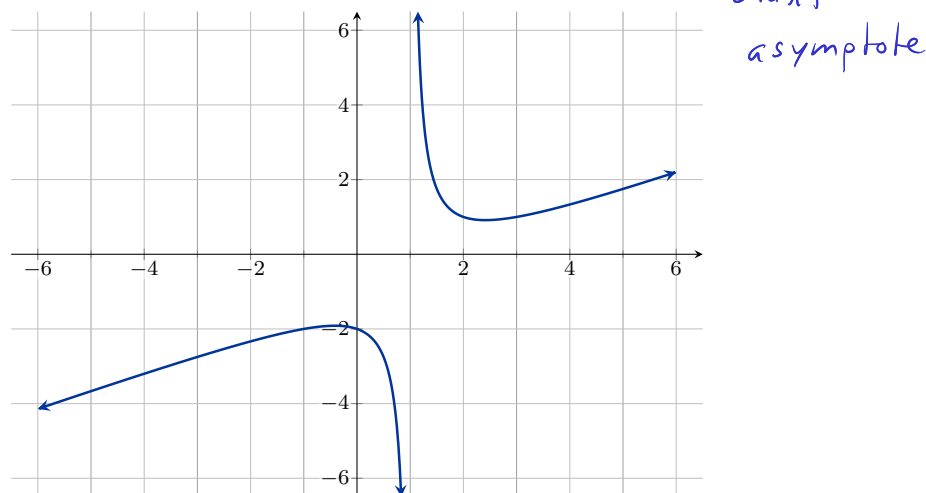
Horizontal asymptote: $y = \frac{2}{1} = 2$

Domain: $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$

Example. Describe the asymptotes, intercepts, and domain of

$$f(x) = \frac{x^2 - 3x + 4}{2(x - 1)} = \underbrace{\frac{1}{2}x - 1}_{\text{slant asymptote}} + \frac{1}{x-1}$$

using the graph below:



$$x \rightarrow 1^-, f(x) \rightarrow -\infty$$

Vertical asymptotes: $x = 1$

$$x \rightarrow 1^+, f(x) \rightarrow \infty$$

$$\frac{2x^4 - 2x^3 + x^2 + 2x - 2}{x^4 - 1} \left(\frac{\frac{1}{x^4}}{\frac{1}{x^4}} \right) = \frac{2 - \frac{2}{x} + \frac{1}{x^2} + \frac{2}{x^3} - \frac{2}{x^4}}{1 - \frac{1}{x^4}}$$

When the degree of the numerator is one more than the degree of the denominator, we have a slant asymptote

Horizontal asymptote: N/A

Domain: $(-\infty, 1) \cup (1, \infty)$